



Dragon flight a success

SpaceX's Dragon spacecraft makes a successful launch in a mission to resupply the International Space Station <http://dgit.in/DragonSuccess>



Saturn's 63rd moon

Peggy, the latest body that could turn out to be Saturn's moon may disintegrate if it continues along current path <http://dgit.in/SaturnPeggy>

Space Age

of the ship facing the direction of travel with the rest being towed behind it. This design concept is traced back to Robert Goddard's early designs for a rocket and is surprisingly effective for interstellar travel. The ship is designed to be launched from space powered by orbital lasers which is a more fuel efficient means of propulsion than using stored fuel banks for thrust.

The lasers power the photon sails of the craft and avoid damaging the ship thanks to a mirror shield on the ships rear. Based on the movie's design, the 1.5 kilometer long ship has photon sails with a diameter of 16 kilometers, capable of accelerating the ship at 1.5 g for nearly half a year and then being retracted into the craft. This push allows the ship to achieve a velocity of 210,000 kilometers per second or 70 percent the speed of light until it reaches its destination. As it arrives at its destination the yet to be invented hybrid fusion and matter-antimatter engines in front of the vessel burn for nearly half a year to decelerate the ship to zero velocity.

It is an elegant design where the bulk of the ship's structure is dedicated to propulsion with a relatively smaller volume devoted to cargo, habitation and crew areas. The Venture Star is essentially a small space station attached to a massively powerful engine capable of interstellar flight. It is an efficient, cost effective and reusable design that allows for maximum returns on investment. But not so much in terms of adventure - realism has its cost.



The Venture Star with its realistic propulsion system – engine hauling the rest of the ship

Other ship designs that push us closer to the exotic include the biomechanical starships from the TV show Farscape called the Leviathans. These epic class of interstellar spacecraft although thoroughly outside the realm of reality (for now) do offer useful insights. A spacecraft that is

"born" in space rather than created would offer a host of benefits – an internal ecosystem designed to produce usable resources like water and food, the ability to respond at the speed of quantum-bionics and adaptable genomic traits that could support the needs of the crew.

The Leviathan helmed by the heros of Farscape is called Moya and was a more than a ship – it was a being in its own right that worked in symbiosis with its crew on their journey through space. Although the field of biomechanics and biocomputers is in its infancy there is no reason that in a 100 years, if not sentience, we would have a spacecraft designed with all the biological elements from the show if not more.

Already using nano-bionics plants are being hot-wired to create more energy. Extrapolate this and you could power a biomechanical spacecraft infinitely, as long as it was sufficiently exposed to photons in space. Such a spacecraft would embody the idea of life in space far more than any other we can imagine today.

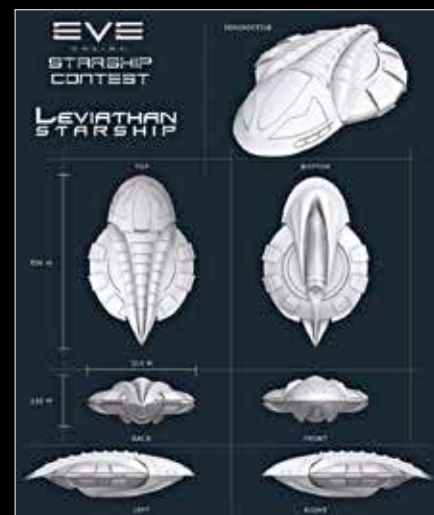
And yet the biggest problem of interstellar space travel remains propulsion - the ability to generate the massive amounts of energy needed to get us to velocities that wouldn't make the universe seem so large. But that too is a changing reality.

Not Warp Drive - Alcubierre Drive!

If ever an astronomer in the future was to look across the galaxy at Alpha Centauri



and see signs of human intelligence shining back at Earth it would most likely be thanks to spacecrafts powered by an Alcubierre drive. This form of theoretical propulsion allows a spacecraft to travel faster than the speed of light by playing around with the very fabric of spacetime -



SPOILERS! Moya, the Leviathan from Farscape gave birth to a son - Talyn.

and it is real - sort of. The theory proposed by Alcubierre allows for the contracting of space in front of a spacecraft and the expanding of space behind it - to literally alter the spacetime continuum to achieve faster than light velocities.

Miguel Alcubierre, who proved this possibility using Einstein's field equations in 1994, arrived at a solution where instead of a spacecraft moving at light speed, it was space that curved around it. The idea of the Alcubierre drive works within Einstein's theory of general relativity which normally doesn't allow for anything to move faster than the speed of light. Such a spacecraft would then be static within a "warp bubble" which would be riding along the spacetime curvature as the Alcubierre drive warped the space around it.

The design of such a ship was a centerpiece discussion at the 100 Year Starship program in 2012. Scientists predicted that such a drive could be possible at substantially less power requirements than previously thought and even laid out a basic design. The Alcubierre drive would require the spaceship to be shaped like a tapered oval in the form of a prolate spheroid attached to a large encircling donut like ring. The donut would be composed of exotic matter that would be controlled to cause space-time to curve around the starship as it was suspended in a warp bubble. This concept if implemented with enough exotic matter and technological innovation,